

I joined
Post Invoice Deduction
Row count jumped
totals quietly shifted

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

DAY 9 / 15

04/08/26

- Yesterday : Created 4 CTE for readability by adding 4 JOINS deep
- Joined pre invoice deduction on customer code and fiscal year
- Finance team wants one more layer **discount_pct** and **other_deductions_pct**

The Trap ❌

- Joined post invoice the same way - ON customer_code AND fiscal_year

Row counts Jumped

Totals quietly inflated



Requirment

DAY 9 / 15

04/08/26

match the key to the deductions table



The Fix

Sales report > scope the raw sales here

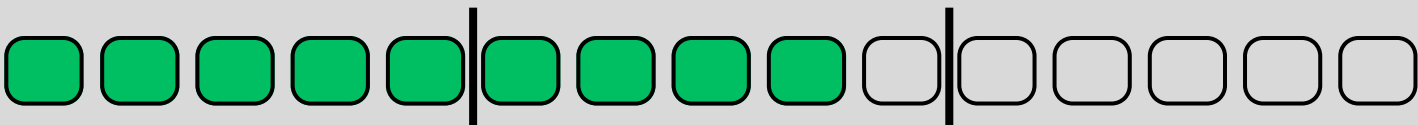
Gross sales report > attach the price , sum to gross

pre invoice report > attach the deductions %

net invoice sales report > attach the deductions %

post invoice report > attach the deductions %

final SELECT > JOIN the final key . true net sales calculation



Bonus Find

DAY 9 / 15

04/08/26

JOIN key is safe when you use match the deductions table

customer_code + fiscal_year

Pre invoice

----- incorrect calculation when joining post invoice

customer_code + fiscal_year + date

+ Post Invoice

----- correct calculation



A Quick Analogy

DAY 9 / 15

04/08/26

Billing

Pickup

Every shipment logged as booked (sales_report)

Rate card applied

price attached per shipment (gross_sales_report)

Loyalty discount

one yearly per customer (pre_invoice_report)

Route discount

shipment cut, own tracking id (post_invoice_report)

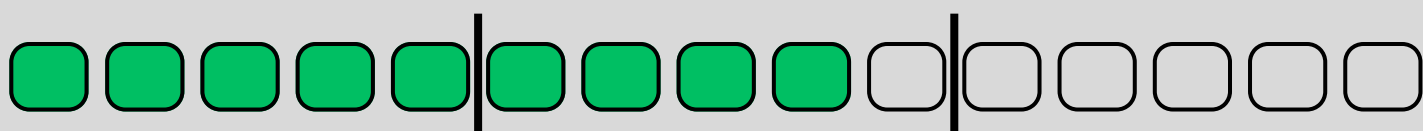
Final invoice

clean bill , true net charge (final SELECT)

One direction , One job per stage



```
WITH sales_report AS (  
  SELECT fsm.date, fsm.fiscal_year, fsm.product_code,  
         fsm.customer_code, fsm.sold_quantity  
  FROM fact_sales_monthly fsm ),  
  
gross_sales_report AS (  
  SELECT  
    sr.date , sr.customer_code, sr.fiscal_year, sr.product_code,  
    sr.sold_quantity * fgp.gross_price gross_sales_amount  
  FROM sales_report sr  
  JOIN fact_gross_price fgp  
  ON sr.product_code = fgp.product_code  
  AND sr.fiscal_year = fgp.fiscal_year ),  
  
pre_invoice_report AS (  
  SELECT  
    preivd.customer_code, preivd.fiscal_year,  
    preivd.pre_invoice_discount_pct  
  FROM fact_pre_invoice_deductions preivd),
```



```
net_invoice_sales_report AS(
SELECT gsr.date ,gsr.fiscal_year, gsr.product_code , gsr.customer_code,
gsr.gross_sales_amount,
gsr.gross_sales_amount *
(1 - pir.pre_invoice_discount_pct) AS net_invoice_sales_amount
FROM gross_sales_report gsr
JOIN pre_invoice_report pir
ON pir.customer_code = gsr.customer_code
AND pir.fiscal_year = gsr.fiscal_year),

post_invoice_report AS (
SELECT fpid.date , fpid.customer_code , fpid.product_code,
fpid.discounts_pct , fpid.other_deductions_pct
FROM fact_post_invoice_deductions fpid )
```




```
SELECT nistr.fiscal_year , nistr.customer_code,  
ROUND(nistr.gross_sales_amount,2) gross_sales_amount,  
ROUND(nistr.net_invoice_sales_amount,2) net_invoice_amount,  
ROUND(nistr.net_invoice_sales_amount  
        * (1 - (poir.discounts_pct + poir.other_deductions_pct)), 2) AS  
net_sales_amount  
FROM net_invoice_sales_report nistr  
JOIN post_invoice_report poir  
ON poir.customer_code = nistr.customer_code  
AND poir.product_code = nistr.product_code  
AND poir.date = nistr.date
```



Results

DAY 9 / 15

04/08/26

fact_sales_monthly
fact_gross_price
dim_product
fact_pre_invoice_deductions
Day 8
fact_post_invoice_deductions
SQL File 8* x

Limit to 50000 rows

```

35
36 SELECT nistr.fiscal_year , nistr.customer_code,
37 ROUND(nistr.gross_sales_amount,2) gross_sales_amount,
38 ROUND(nistr.net_invoice_sales_amount,2) net_invoice_amount,
39 ROUND(nistr.net_invoice_sales_amount
40      * (1 - (poir.discounts_pct + poir.other_deductions_pct)), 2) AS net_sales_amount
41 FROM net_invoice_sales_report nistr
42 JOIN post_invoice_report poir
43 ON poir.customer_code = nistr.customer_code
44 AND poir.product_code = nistr.product_code
45 AND poir.date = nistr.date
46

```

Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Fetch rows:

	fiscal_year	customer_code	gross_sales_amount	net_invoice_amount	net_sales_amount
▶	2022	80007195	958974.58	678090.92	377832.26
	2022	80007195	861575.37	609219.94	402877.15
	2022	80007196	856008.81	614443.12	391215.94
	2022	80007196	811769.24	582687.96	360217.70
	2022	80007195	797312.26	563779.50	367189.59
	2022	80007196	754426.79	541527.55	309320.54
	2022	80007195	714754.70	505403.04	311227.20
	2022	80007196	699228.71	501906.37	313892.24

Result 10 x

Output

Action Output

#	Time	Action	Message
1	18:13:36	WITH sales_report AS (SELECT fsm.date, fsm.fiscal_year,fsm.product_code, fsm.customer_code, fsm.sold_quantity FR...	50 row(s) returned
2	18:18:31	WITH sales_report AS (SELECT fsm.date, fsm.fiscal_year,fsm.product_code, fsm.customer_code, fsm.sold_quantity FR...	1424923 row(s) returned



DAY 9 / 15

04/08/26



I Vishnu Ram Sachin I Data Analyst

SQL THROUGH ONE DATASET CHALLENGE